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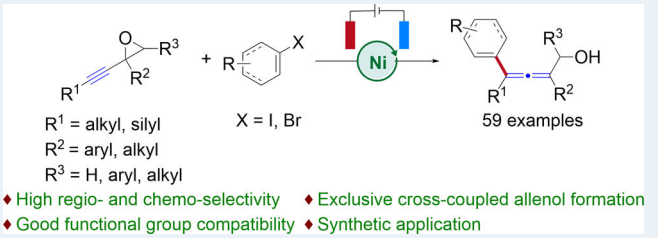
Electronickel-Catalyzed 1,5-Hydroarylation of Alkynyl Epoxides

Wenyao Li, Hao Yu,* Hui Qian,* and Shengming Ma*

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ABSTRACT: Electrochemical transformations of epoxides have emerged as a useful platform for alcohol synthesis, and their reactivity and selectivity are strongly dictated by the substitutions of the epoxides. Incorporation of an alkynyl substituent into the epoxide framework offers a promising yet unexplored avenue toward highly functionalized different alcohol derivatives posing formidable challenges in selectivity control. Here, we report an earth-abundant metal of nickel-catalyzed cross-electrophile coupling between readily available alkynyl epoxides and aryl/alkenyl halides, enabling the highly selective synthesis of allenols under mild electrochemical conditions. This protocol exhibits a broad substrate scope, preferential C−O bond activation over C−X bonds, and a decent chemoselectivity for C−I activation over the C−Br bond, while tolerating a wide array of synthetically valuable functional groups, such as free hydroxyl, ketone, carboxylate, methoxy, TBS ether, *N*-Boc, highly strained cyclopropyl, C═C bonds, C−TMS, C−F bonds, C−Cl bonds, C−Br bonds, and even C−B bonds. Moreover, this methodology may also apply to the synthesis of allenols bearing pharmaceutical and biologically active moieties, and the downstream derivatizations of allenol products have also been realized, offering a modular platform for the synthesis of diversified functional structures.



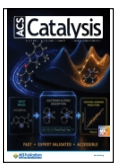
KEYWORDS: *alkynyl epoxides, electrosynthesis, nickel catalysis, allenol, cross-electrophile coupling*

INTRODUCTION

Alcohols are ubiquitous in pharmaceuticals, agrochemicals, fragrances, and flavorings and also serve as indispensable synthetic intermediates in both laboratory and industrial settings.[1,2] Accordingly, the development of efficient and selective synthetic strategies for functionalized alcohols remains a subject of enduring interest and practical importance.[3] In organic chemistry, epoxides occupy a privileged position owing to their easy preparation and intrinsic ring strain, which renders them susceptible to reductive or nucleophilic ring-opening reactions.[4] On the other hand, organic electrosynthesis has re-emerged as a powerful and sustainable platform for redox transformations, leveraging electrons as traceless reagents to generate reactive intermediates under mild and environmentally benign conditions.[5] Electrochemical transformations of epoxides have also started to attract attention: Lu and co-workers disclosed the electroreduction of epoxides with carefully tuned proton sources suppressing hydrogen evolution to enable regioselective access to Markovnikov alcohols from alkyl epoxides (Scheme 1A-i) and anti-Markovnikov alcohols from aryl or benzyl epoxides (Scheme 1A-ii).[6] Qiu et al. and Zhang et al. independently achieved electrocarboxylation of aryl epoxide with carbon dioxide, affording β-hydroxy carboxylic acids with a very high selectivity (Scheme 1A-iii).[7] These studies collectively underscore the decisive influence of the substitution on the reactivity and selectivity of epoxides.[8]

We envisioned that the incorporation of an alkynyl moiety could unlock new reactivity patterns of epoxides. The first thing that came to our mind is their two-component coupling reactions with organic halides, which may potentially furnish a new methodology for the syntheses of different functionalized two-component alcohols, i.e., allenols **P1**, homopropargylic alcohols **P2**, or propargylic alcohols **P3** (Scheme 1B, top). This design is accompanied by formidable challenges (Scheme 1B, bottom): (i) the simultaneous presence of two strained C−O bonds raises the challenge of competing cleavage of two distinct C−O bonds; (ii) the incorporation of an alkyne substituent complicates the regioselectivity, owing to the presence of multiple reactive sites that can engage in coupling with aryl halides;[9] (iii) direct electroreduction of the alkynyl epoxide represents a facile but undesired pathway forming mono-component allenols **BP1**, homopropargylic alcohols **BP2**, and propargylic alcohols **BP3** as reported in the literature;[6] (iv) possible allenols **P1** or propargyl alcohols **P3** are susceptible to over-reduction[10] or cyclization[11] under electrochemical conditions.

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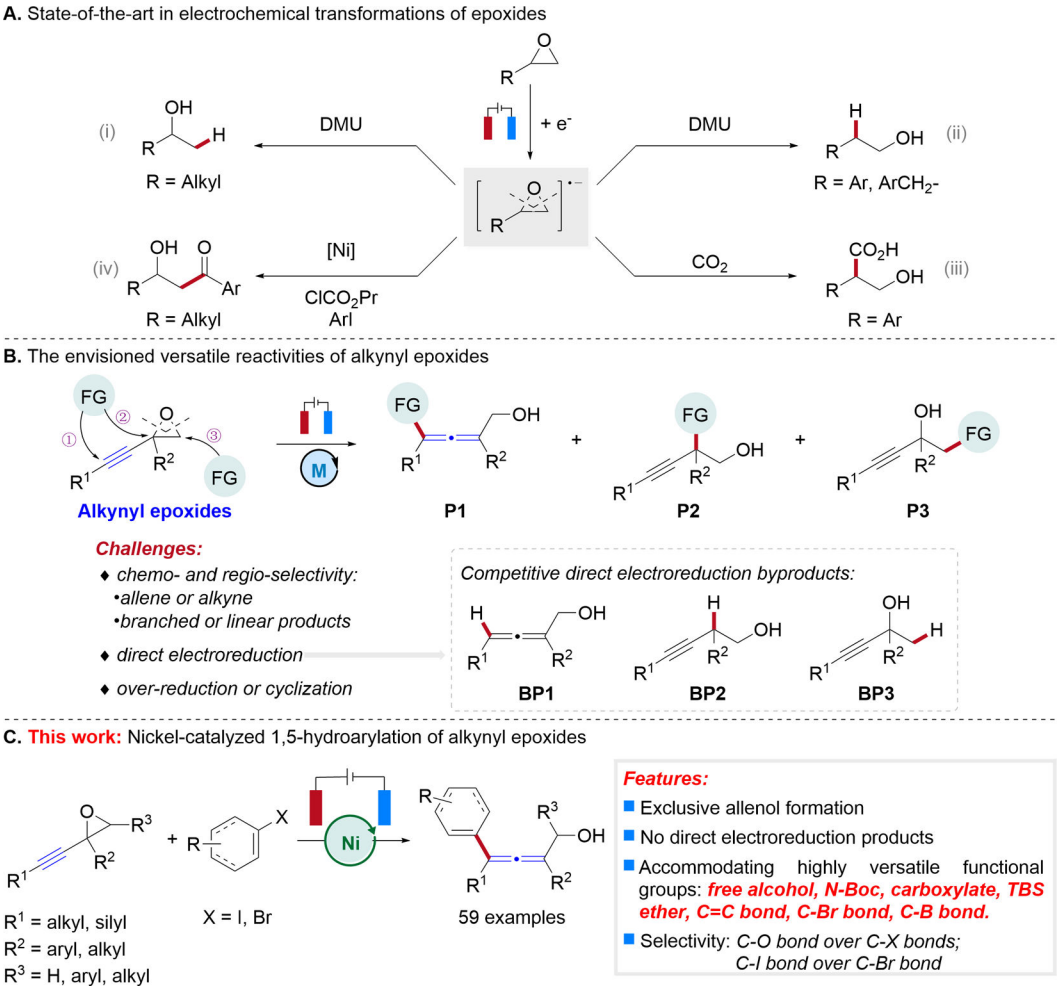


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Scheme 1. (A–C) Electrochemical Transformation of Differently Substituted Epoxides



We reasoned that merging transition metal catalysis with electrochemistry could overcome these obstacles by affording greater control over bond activation, regioselectivity, and product distribution.[12,13] Such a merger offers complementary advantages: electrochemistry provides a tunable and sustainable means to generate reactive intermediates, while transition metal catalysis governs their selective capture and subsequent transformation. The synergy between these two activation modes can orchestrate complex redox sequences that are difficult to achieve by either approach alone. In this context, Shi et al. demonstrated an elegant nickel-catalyzed electroreductive coupling of alkyl epoxides with aryl halides in the presence of isopropyl chloroformate as a CO surrogate (Scheme 1A-iv).[14] Herein, we report the first example of nickel-catalyzed electrochemical two-component coupling of alkynyl epoxides with organic halides (Scheme 1C). The current protocol enables efficient and exclusive synthesis of structurally diverse allenols,[15] a very important class of compounds accommodating an array of synthetically reactive groups without protection and deprotection.

RESULTS AND DISCUSSION

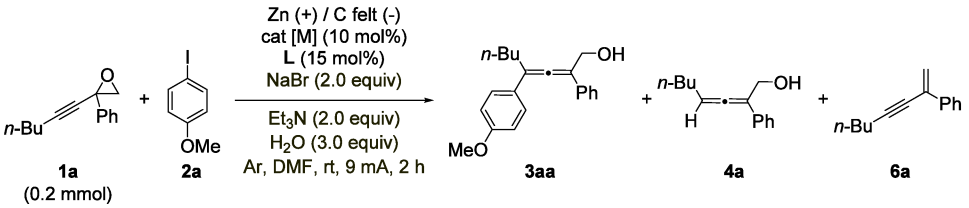
We first investigated the reaction of alkynyl epoxide **1a** and 4-methoxyphenyl iodide **2a** in DMF under a constant current of 9 mA in DMF (Table 1). When $Pd(dba)_2$ was employed as the catalyst in combination with BINAP, no cross-coupled product was formed with NaBr as the electrolyte together with 2.0 equiv of Et_3N and 3.0 equiv of H_2O (Table 1, entry 1). It is interesting to observe that with DPEphos, DPPB, or BIPHEP as the ligand (Table 1, entries 2–4), the reaction of **1a** alone afforded 2-phenyl-2,3-heptadienol **4a** and enyne product **6a**, indicating the cleavage of the propargylic C–O bond in **1a**. When $FeCl_3$, $FeCl_2$, or $CoCl_2$ was used with bipy as the ligand, similar reactivity was observed (Table 1, entries 5–7). Interestingly, when $NiCl_2$·DME was applied, the expected cross coupling product of tetra-substituted allenol **3aa** was obtained in 71% yield with 11% of enyne **6a** (Table 1, entry 8).

A decreased yield was observed when other nickel catalysts, such as $NiBr_2$ and $NiBr_2$·DME, were used instead of $NiCl_2$·DME (Table 2, entries 2 and 3). In the absence of nickel catalyst, no coupling product was obtained (Table 2, entry 4), confirming the essential role of nickel in this transformation. Bipy proved to be the most effective ligand, as compared with dtbbpy and 1,10-phen (Table 2, entries 5 and 6). Among the solvents examined, DMF gave the highest efficiency, outperforming DMA and DMSO (Table 2, entries 7 and 8). Other electrolytes including NaI and $TBABF_4$ resulted in a poor reactivity (Table 2, entries 9 and 10). In comparison, when the zinc plate anode was replaced with an aluminum plate, **3aa** was obtained in 49% yield (Table 2, entry 11), while no product was obtained when carbon felt was used (Table 2, entry 12). To our delight, upon fine-tuning the reaction parameters by increasing the electrode surface area and reaction time while reducing the equivalents of aryl iodide and triethylamine, the yield

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Table 1. Catalyst Screening[a]



entry	cat. [M]	ligand	yield of **3aa** (%)[b]	yield of **4a** (%)[b]	yield of **6a** (%)[b]
1	$Pd(dba)_2$	BINAP	0	0	0
2	$Pd(dba)_2$	DPEphos	0	12	4
3	$Pd(dba)_2$	DPPB	0	22	15
4	$Pd(dba)_2$	BIPHEP	0	27	4
5	$FeCl_3$	bipy	0	26	7
6	$FeCl_2$	bipy	0	28	12
7	$CoCl_2$	bipy	0	23	8
8	$NiCl_2$·DME	bipy	71	0	11

[a]Conditions: undivided cell, constant current = 9 mA, Zn plate anode (25 × 10 × 0.2 mm^3), C felt cathode (25 × 10 × 5 mm^3), **1a** (0.2 mmol), **2a** (0.6 mmol), cat [M] (10 mol%), **L** (15 mol%), Et_3N (0.4 mmol), H_2O (0.6 mmol), NaBr (0.4 mmol), DMF (4 mL), Ar atmosphere, room temperature, 2 h. [b]The yield was determined by ^1H NMR analysis using dibromomethane as the internal standard.

was improved to 77% (74% isolated) when the reaction was scaled up to 1.0 mmol (Table 2, entry 13). Adjusting the current to either 8 or 10 mA slightly reduced the yield (Table 2, entries 14 and 15). No product formation was observed when the reaction was conducted with Zn or Mn powder to provide the electron, or without electricity, underscoring the critical role of tunable electrons provided by electricity in driving this process (Table 2, entries 16−18). We also carried out the reaction of **1a** with **2a** under the standard conditions with an IKA Electrasyn instrument, which afforded the allenol **3aa** in a similar yield of 71% yield (Table 2, entry 19 vs entry 13).

With the optimized conditions in hand, we first evaluated the substrate scope of aryl halides, and the structures of the resulting allenol products were unambiguously confirmed through the X-ray single-crystal analysis of allenol **3ka** (Scheme 2). Both electron-donating and electron-withdrawing groups at the *ortho*- (**3ac** and **3af**), *meta*- (**3ab**, **3ae**, **3bg**, and **3ah**), and *para*-positions (**3aa**, **3ad**, **3aj**, **3ak**, **3cl**, **3cm**, **3an**, **3ao**, **3bp**, **3bq**, **3ar**, **3as**, and **3bt**) of iodobenzenes are compatible, yielding the corresponding α-allenols in moderate to good yields exclusively. Steric effects play a role since the *ortho*-substituted aryl iodides provided lower yields (**3ac** vs **3aa**, **3ab**; **3af** vs **3ad**, **3ae**). A wide array of functional groups, such as alkoxyl (**3ac** and **3ca**), phenyl (**3ak**), chloride (**3ar**), trifluoromethyl (**3as**), nitrile (**3bt**), naphthyl (**3bu**), and pendant benzoheterocycles (**3av** and **3aw**), were well-tolerated under the current conditions. Notably, substrates with reactive functional groups such as hydroxyl (**3an**), Boc-protected amine (**3ao**), carboxylate (**3bp**), and bromide (**3ad**) all delivered the α-allenols in 59−67% yields, underscoring the excellent scope of this strategy.

To further probe chemoselectivity, bifunctional substrate **2q** bearing both an electrophilic C−I bond and a nucleophilic C−B bond was evaluated, affording allenol **3bq** in 50% yield. This outcome highlights the strong preference of those electrochemical conditions for cross-electrophile coupling over electrophile−nucleophile pathways. Gratifyingly, the substrate scope was further extended to heteroaryl halides. 7-Iodoquinoline (**2x**) as well as 3-iodopyridines (**2y** and **2z**) participated smoothly, delivering allenols **3ax**−**3az** in good yields. In addition to aryl iodides, aryl bromides also performed well, affording α-allenols **3aA**−**3aD** smoothly when the electrolyte was switched from NaBr to NaI. To facilitate product purification, the reactions of **2A** and **2B** were carried out on a 0.2 mmol scale under a constant current of 2 mA. Notably, 3-bromoquinoline was also successfully engaged to furnish product **3aD**. Attempts to further extend the scope of aryl halide substrates to aryl chlorides proved unsuccessful, as neither *p*-methoxy nor *p*-methoxycarbonyl-substituted phenyl chloride afforded the allenol product (Table S5). However, particularly noteworthy is the selective reactivity observed with bromophenyl iodides, where the transformation occurred exclusively at the C−I bond to furnish the corresponding allenols (**3ad**−**3af**) while leaving the C−Br bond untouched. Such chemoselective activation of aryl iodides over the C−Br bond in the same molecule represents a valuable feature for further stepwise elaboration, enabling the streamlined synthesis of aryl bromide-flanked allenol scaffolds that can serve as versatile intermediates for further cross-coupling or functionalization transformations.

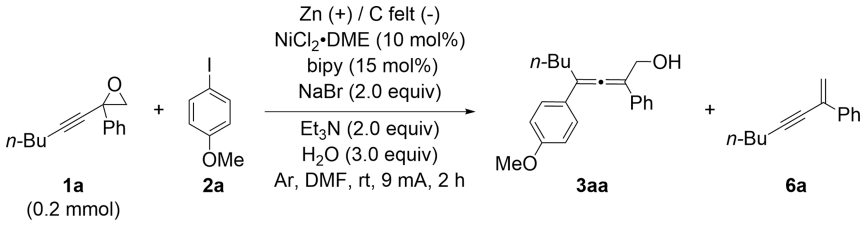
The generality of the reaction was further extended to vinyl bromides, enabling the synthesis of alkenyl allenes: Cyclic vinyl bromides, including cyclopentenyl (**3aE**) and cyclohexenyl derivatives (**3aF**), as well as benzene-fused analogues (**3aG** and **3aH**), underwent smooth conversion under the standard conditions.

Next, differently substituted alkynyl epoxides **1** were examined. The propargylic substituent R^2 could be aryl (**3da** and **3ea**), alkyl (**3fa** and **3ga**), and piperidinyl (**3ha**), affording products in moderate yields. The alkynyl epoxide bearing a 4-bromophenyl moiety (**3da**) underwent exclusive ring opening at the epoxide site, leaving the aryl bromide functionality untouched. The C−C triple bond substituent (R^1) can accommodate primary alkyl chains of varying lengths (**3ia**, **3ja**, **3da**, and **3ea**) and primary alkyl groups bearing functional groups such as chloro (**3ga**), phenyl (**3ka**), and TBS ether (**3la**). Moreover, secondary alkyl groups including the highly strained cyclopropyl (**3ma**) and piperidinyl (**3na**), as well as the trimethylsilyl group (**3oa**), are also well-tolerated. Our attempts to utilize alkynyl epoxides bearing a phenyl on the terminal position

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Table 2. Screening of the Reaction Conditions[a]



entry	variations	yield of **3aa** (%)[b]	yield of **6a** (%)[b]
1	none	71	11
2	$NiBr_2$ instead of $NiCl_2$·DME	57	18
3	$NiBr_2$·DME instead of $NiCl_2$·DME	64	5
4[c]	no $NiCl_2$·DME	0	6
5	dtbbpy instead of bipy	40	17
6	1,10-phen instead of bipy	24	20
7	DMA instead of DMF	56	6
8	DMSO instead of DMF	54	8
9	NaI instead of NaBr	61	16
10	$TBABF_4$ instead of NaBr	30	9
11	Al (+) instead of Zn (+)	49	3
12	C felt (+) instead of Zn (+)	0	0
13[d]	1.0 mmol, 9 mA	77(74)[e]	8
14[d]	1.0 mmol, 8 mA	74	8
15[d]	1.0 mmol, 10 mA	75	6
16	Mn powder (3.0 equiv) instead of electricity	0	0
17	Zn powder (3.0 equiv) instead of electricity	0	0
18	no electricity	0	0
19[d]	IKA Electrasyn	71	8

[a]Conditions: undivided cell, constant current = 9 mA, Zn plate anode (25 × 10 × 0.2 mm^3), C felt cathode (25 × 10 × 5 mm^3), **1a** (0.2 mmol), **2a** (0.6 mmol), $NiCl_2$·DME (10 mol%), bipy (15 mol%), Et_3N (0.4 mmol), H_2O (0.6 mmol), NaBr (0.4 mmol), DMF (4 mL), Ar atmosphere, room temperature, 2 h. [b]The yield was determined by ^1H NMR analysis using dibromomethane as the internal standard. [c]A 9% NMR yield of **4a** was detected. [d]The reaction conditions of 1 mmol scale: undivided cell, constant current = 9 mA, Zn plate anode (40 × 15 × 0.2 mm^3), C felt cathode (40 × 15 × 5 mm^3), **1a** (1.0 mmol), **2a** (2.75 mmol), $NiCl_2$·DME (10 mol%), bipy (15 mol%), Et_3N (1.5 mmol), H_2O (3.0 mmol), NaBr (2.0 mmol), DMF (20 mL), Ar atmosphere, room temperature, 12 h. [e]The yield in parentheses indicates isolated yield.

of the C−C triple bond were unsuccessful, as the reactions afforded low NMR yields and the products could not be efficiently purified. Notably, this transformation was not limited to 1,1-disubstituted alkynyl epoxides. 1,1,2-Trisubstituted epoxides also underwent this transformation efficiently, allowing the formation of secondary allenols **3pa**, **3qa**, **3ra**, and **3sa** in 32−76% yields. The stereochemical assignment of allenol **3pa** was established through NOE measurement on its dihydrofuran derivative obtained by treating (1*R**,3R_a*)-**3pa** with $AgNO_3$ (Scheme 3D; see details in the Supporting Information, pages S65−S66).[16,17] Those of **3qa** and **3ra** were assigned analogously (Scheme 3E,F; see details in the Supporting Information, pages S66−S71).

The reaction exhibits several substrate limitations (Figure S5). Aryl bromides bearing nitro, acyl chloride, or cyclic amide groups and acyclic alkenyl iodides were not compatible under the standard reaction conditions. Regarding alkynyl epoxides, substrates with a bulky *tert*-butyl group at the propargylic position and terminal alkyne substrate were also unsuitable for this transformation.

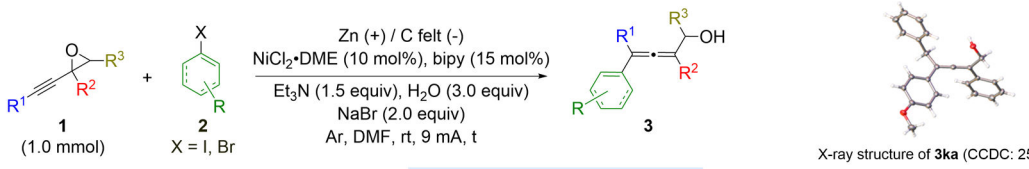
Moreover, this cross-coupling reaction could also be applied to the synthesis of complicated allenols containing bioactive or complex fragments derived from drug molecules and natural products. As shown in Scheme 3A, various delicate aryl iodides such as the derivatives of oxaprozin (**3aJ**), isoxepac (**3aK**), gemfibrozil (**3aL**), indomethacin (**3aM**), probenecid (**3bN**), herbicide 2,4-D (**3bO**), geraniol (**3bP**), and phenothiazine (**3bQ**) were proven to be effective coupling partners, yielding the corresponding α-allenols with decent yields and a high regioselectivity. During our investigation of the substrate scope, most cross-coupled allenol products were accompanied by the formation of enyne byproducts, which were generated exclusively from the alkynyl epoxide substrates.

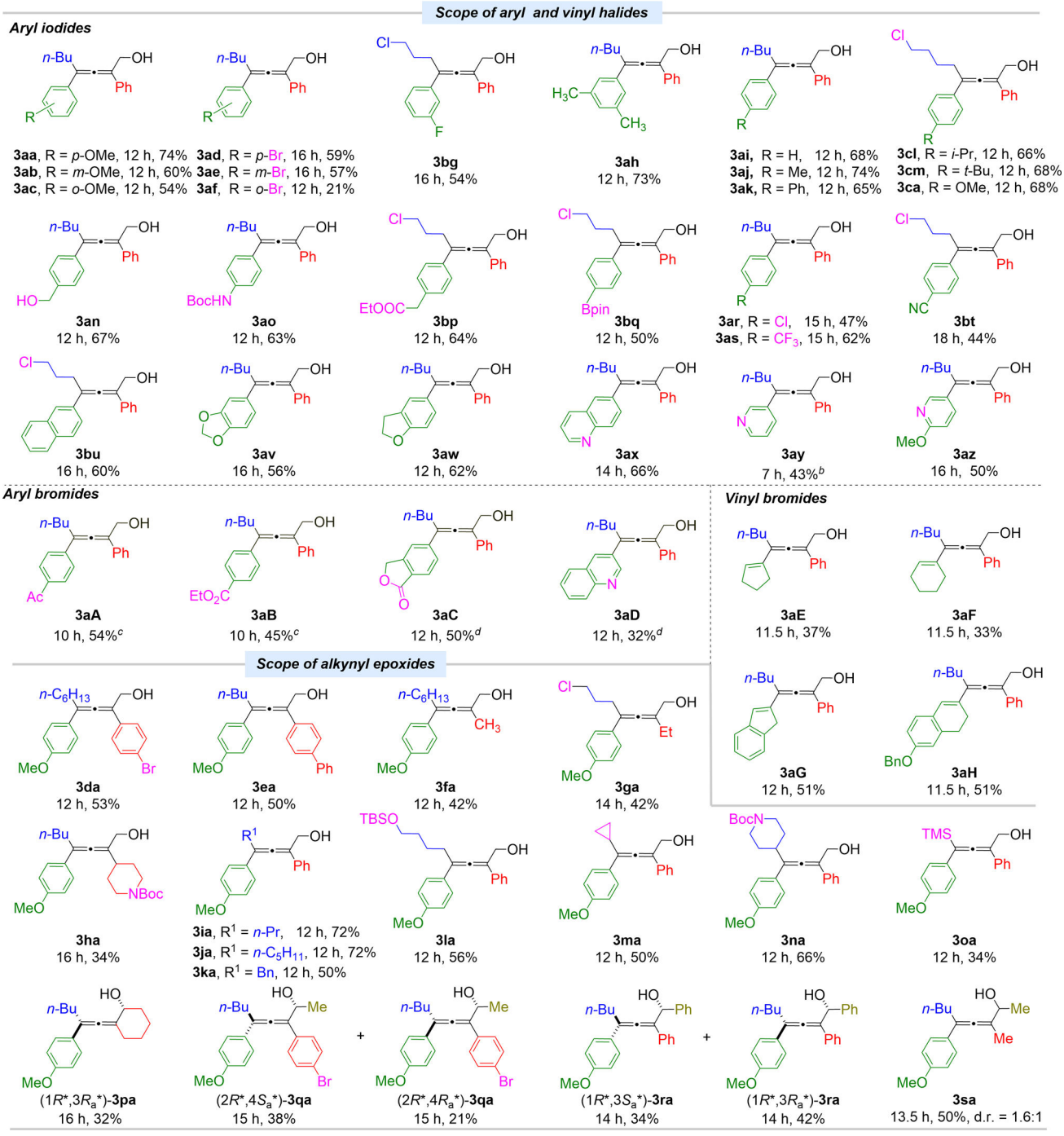
To further demonstrate the practicality of this nickel-electrocatalyzed protocol, the electrochemical cross-coupling of 5 mmol of alkynyl epoxide **1a** and 4-methoxyphenyl iodide **2a** was conducted under constant current electrolysis (45 mA) to provide the desired allenol product **3aa** (0.97 g, 63%) (Scheme 3B). Given the synthetic significance of allenols, the downstream transformations have been successfully realized for the synthesis of various important structures (Scheme 3C): In the presence of a catalytic amount of $AgNO_3$, 2,3-allenol **3aa** underwent cycloisomerization to produce the dihydrofuran derivative **10**;[18] iodine-mediated 5-*endo*-*trig* cyclization of **3aa** resulted in the formation of 3-iododihydrofuran product **11** with a quantitative yield;[19] a $PdCl_2$-catalyzed coupling-cyclization reaction of **3aa** in the presence of allyl bromide afforded polysubstituted

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Scheme 2. Scope of Aryl Halides and Alkynyl Epoxides[a]



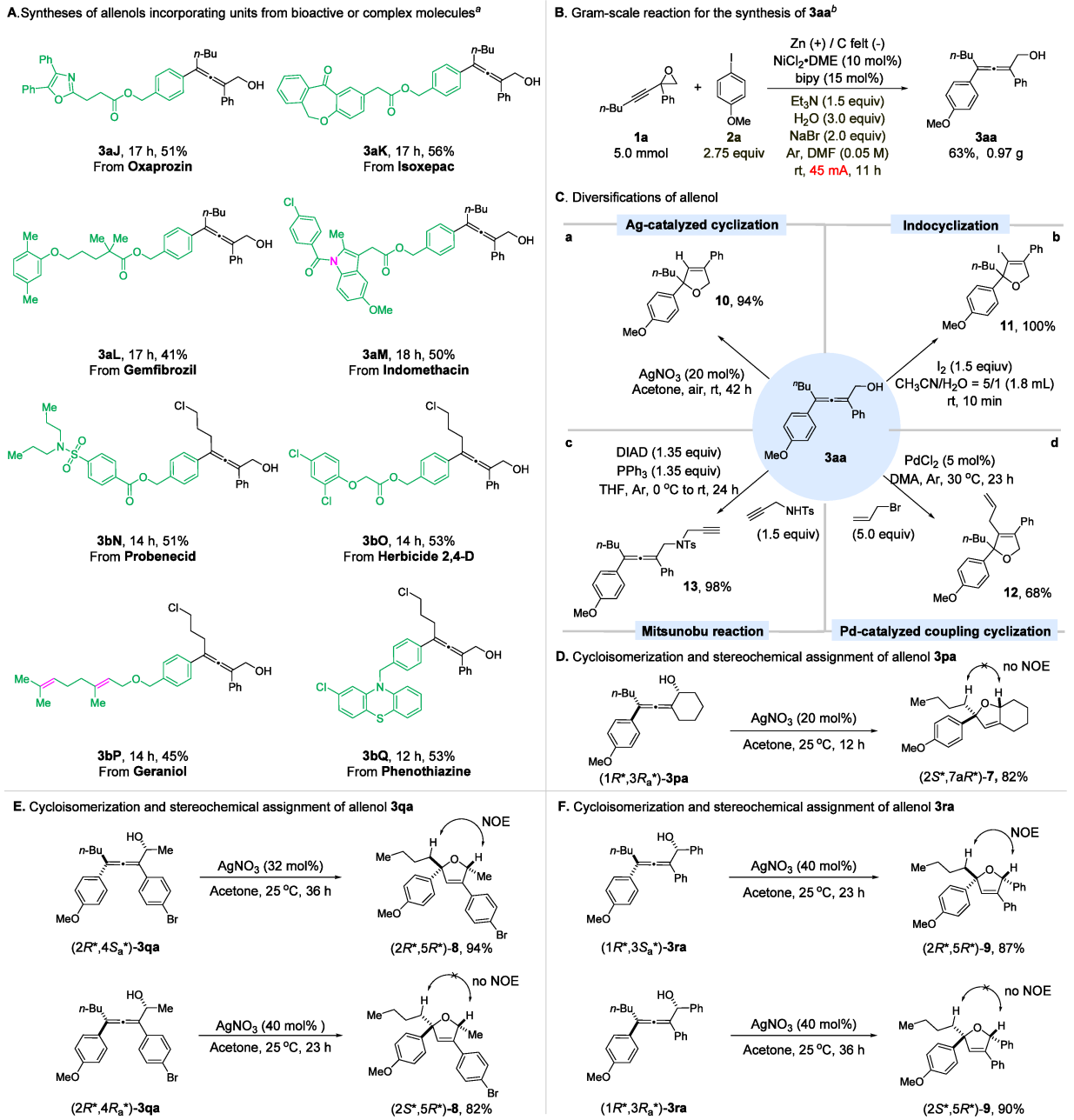


[a]Conditions: undivided cell, constant current = 9 mA, Zn plate anode (40 × 15 × 0.2 mm^3), C felt cathode (40 × 15 × 5 mm^3), epoxide **1** (1.0 mmol), aryl/alkenyl halide **2** (2.75 mmol), $NiCl_2$·DME (10 mol%), bipy (15 mol%), Et_3N (1.5 mmol), H_2O (3.0 mmol), NaBr (2.0 mmol), DMF (20 mL), Ar atmosphere, room temperature. [b]Conditions: undivided cell, constant current = 4 mA, Zn plate anode (25 × 10 × 0.2 mm^3), C felt cathode (25 × 10 × 5 mm^3), **1** (0.2 mmol), **2** (0.6 mmol), $NiCl_2$·DME (10 mol%), bipy (15 mol%), Et_3N (0.4 mmol), H_2O (0.6 mmol), NaBr (0.4 mmol), DMF (4 mL), Ar atmosphere, room temperature. [c]Conditions: undivided cell, constant current = 2 mA, Zn plate anode (25 × 10 × 0.2 mm^3), C felt cathode (25 × 10 × 5 mm^3), **1** (0.2 mmol), **2** (0.6 mmol), $NiCl_2$·DME (10 mol%), bipy (15 mol%), Et_3N (0.4 mmol), H_2O (0.6 mmol), NaI (0.4 mmol), DMF (4 mL), Ar atmosphere, room temperature. [d]NaI (2.0 mmol) was used.

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Scheme 3. (A–F) Syntheses of Complicated Allenols and Synthetic Applications



[a]Unless otherwise noted, all reactions were performed under standard conditions; [b]standard conditions with **1** (5.0 mmol), Zn plate anode (70 × 30 × 0.2 mm^3), C felt cathode (70 × 30 × 5 mm^3), constant current = 45 mA; for experimental details, see the Supporting Information.

2,5-dihydrofuran **12** in 68% yield;[20] lastly, the Mitsunobu reaction would yield allenyl propynyl amide **13** in 98% yield.[21]

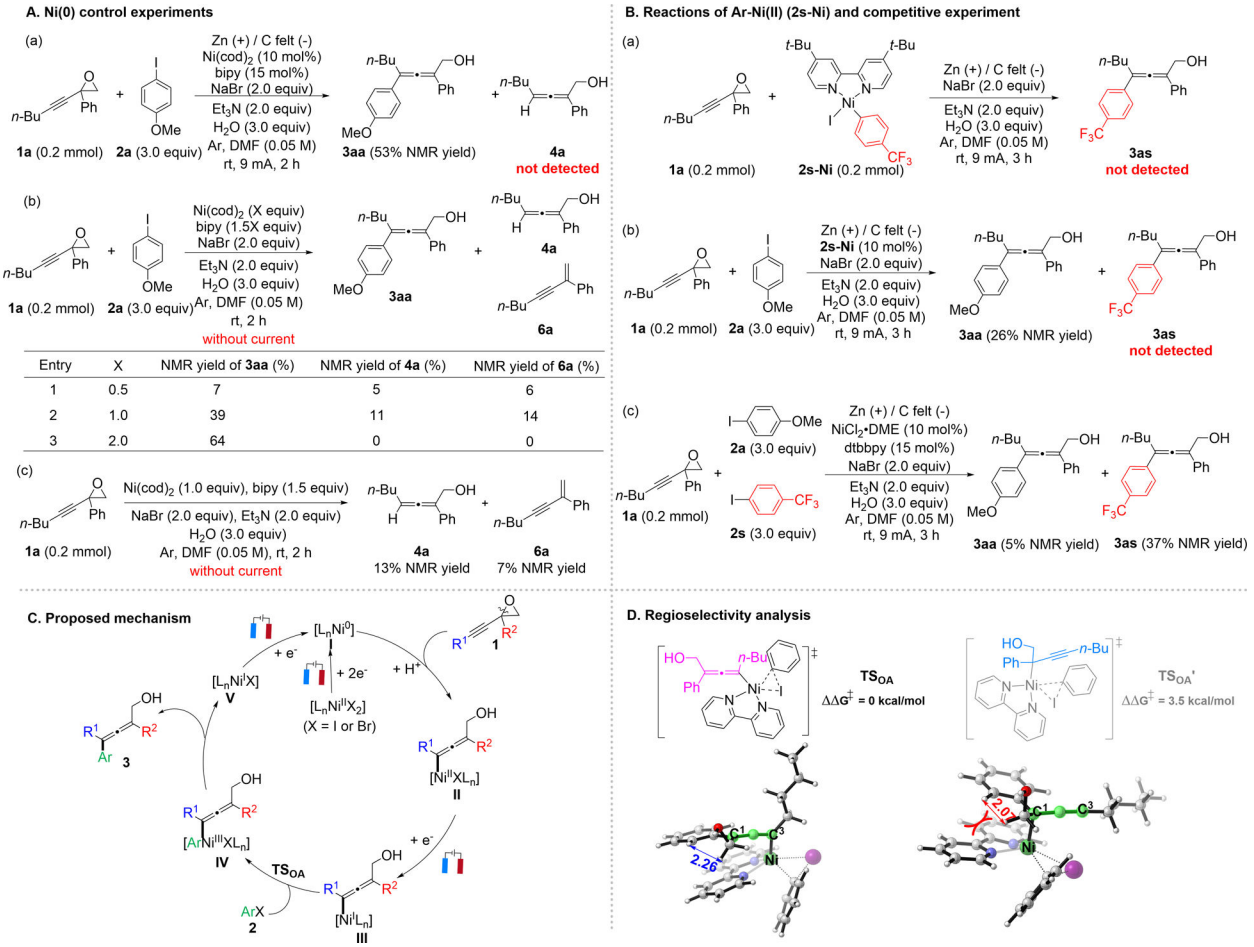
To elucidate the mechanistic details of this electrochemical Ni-catalyzed cross-electrophile coupling, a series of control experiments were conducted. When a catalytic amount of $Ni(cod)_2$ was employed under electrolysis conditions, the allenol product **3aa** was obtained in 53% yield, and no electroreduction product allene **4a** was detected (Scheme 4A-a). The reaction could also proceed under non-electrolytic conditions: treatment with 1.0 equiv of $Ni(cod)_2$ afforded **3aa** in 39% NMR yield together with an 11% NMR yield of **4a** and a 14% NMR yield of **6a**, whereas **3aa** was afforded in 64% yield exclusively upon increasing the nickel loading to 2.0 equiv (Scheme 4A-b, entries 2 and 3). These observations suggest that Ni^0 species may serve as the catalytically active species. In addition, the reaction of alkynyl epoxide **1a** with stoichiometric $Ni(cod)_2$ in the absence of aryl iodide afforded protonated allene **4a** in 13% NMR yield, further confirming that Ni^0 species could react with the alkynyl epoxide to form a reactive organonickel intermediate (Scheme 4A-c).

To clarify the sequence of oxidative addition events in the mechanism, we synthesized a known Ni^{II}-aryl complex **2s−Ni** via oxidative addition of Ni^0 with 4-trifluorophenyl iodide **2s** as reported in ref 22. Stoichiometric reaction between **2s−Ni** and the alkynyl epoxide **1a** yielded no cross-coupled product **3as**, excluding its role as an on-cycle intermediate (Scheme 4B-a). When **2s−Ni** (10 mol %) was used catalytically under the electrochemical

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Scheme 4. (A–D) Mechanistic Studies and Proposed Mechanism



conditions, the reaction of **1a** with 4-methoxyphenyl iodide (**2a**) afforded **3aa** in 26% yield, but no **3as** incorporating the 4-trifluoromethylphenyl group was detected (Scheme 4B-b). Furthermore, in a competition experiment between **2a** and **2s**, the coupling preferentially occurred with 4-trifluorophenyl iodide (Scheme 4B-c). Collectively, these results indicate that **2s**−**Ni** may not be the active intermediate in this reaction pathway.

To probe whether allenyl radicals generated via the reductive activation of alkynyl epoxides **1** are involved in this electrochemical transformation, we conducted additional electrochemical analyses using a three-electrode setup to measure the operating cathodic potentials under different conditions. Under the standard catalytic conditions, the operating cathodic potential was measured to be ∼ −1.5 V vs $Ag/AgNO_3$ (Figure S11). In contrast, in the absence of aryl iodide, nickel catalyst, and bipyridine ligand, under the conditions corresponding to the direct cathodic reduction of the alkynyl epoxide, the cathodic potential was maintained at ∼ −2.4 V vs $Ag/AgNO_3$ (Figure S10). Cyclic voltammogram (CV) experiments revealed that the reduction potential of alkynyl epoxide **1a** is −2.5 V vs $Ag/AgNO_3$, which displayed an obvious positive shift to −2.0 V vs $Ag/AgNO_3$ with the assistance of $ZnCl_2$ (Figure S9). Collectively, these results establish that direct cathodic reduction of the alkynyl epoxide substrate is not possible under our catalytic reaction conditions.

Based on these observations, we propose the mechanistic scenario outlined in Scheme 4C. The catalytically active Ni^0 species **I** is generated via electrochemical reduction of the Ni^{II} precursor. This active Ni^0 complex **I** undergoes oxidative addition with the alkynyl epoxide **1** to afford the allenyl nickel(II) intermediate **II**, which is further reduced at the cathode to form the allenyl Ni^I species **III**. Its subsequent oxidative addition with the aryl halide **2** produces the allenyl aryl Ni^{III} intermediate **IV**, which is followed by reductive elimination to yield the cross-coupled product allenol **3** and the Ni^I species **V**. Cathodic reduction of Ni^I species **V** regenerates Ni^0, completing the catalytic cycle.

Finally, DFT calculations were carried out to rationalize the observed regioselectivity (Scheme 4D).[23] The calculated energy barrier for the oxidative addition leading to allene formation is 3.5 kcal/mol lower than that for the alkyne-forming pathway, indicating a kinetic preference for the allenyl route. Structural analysis of transition states $\mathbf{TS_{OA}}$ and $\mathbf{TS_{OA}}$' revealed that the steric effect governs the pathway selectivity. $\mathbf{TS_{OA}}$' exhibits significant steric congestion, with a short H···H distance (2.07 Å) between hydroxymethyl and phenyl groups on C1. In contrast, $\mathbf{TS_{OA}}$ shows a much less steric repulsion with a longer H···H distance (2.26 Å), accounting for the favored formation of the allenyl product.

■ CONCLUSIONS

In conclusion, we have developed an electro-driven nickel-catalyzed cross-electrophile coupling reaction of alkynyl epoxides with aryl/alkenyl halides, which allows for the efficient construction of diverse α-allenols while tolerating a wide range of synthetically useful functional groups. This method represents

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a compelling alternative to conventional approaches that rely on preformed nucleophiles such as Grignard reagents,[24−26] organozinc species,[27,28] organoboron reagents,[29−32] and terminal alkynes.[33,34] These traditional allenol synthesis suffers from limited functional group tolerance due to the high reactive nature of the organometallic reagents. Although organoboron reagents are relatively stable and less reactive, their tedious preparation using highly reactive reagents further restricts functional group compatibility. This protocol exhibits high regioselectivity and a distinctive selectivity profile, featuring preferential C−O bond activation over C−X bonds as well as pronounced chemoselectivity for C−I activation. Notably, this strategy can be applied to the synthesis of allenols incorporating biologically relevant motifs under mild conditions and demonstrated that allenols can serve as valuable intermediates for further synthetic transformations.

METHODS

Typical Procedure for the Synthesis of Allenol 3aa

To an oven-dried 50 mL Schlenk tube were added bipy (23.8 mg, 0.15 mmol), NaBr (207.9 mg, 2.0 mmol), 4-methoxyiodobenzene (650.3 mg, 2.75 mmol), and $NiCl_2$·DME (21.9 mg, 0.1 mmol) sequentially. The tube was evacuated and backfilled with argon three times using a vacuum line. Then, a solution of **1a** (200.0 mg, 1.0 mmol) in DMF (20 mL), Et_3N (211 μL, d = 0.728 g/mL, 153.6 mg, 1.5 mmol), and H_2O (54 μL, d = 0.998 g/mL, 53.9 mg, 3.0 mmol) were added to the tube. A zinc plate anode (40 × 15 × 0.2 mm^3) and a carbon felt cathode (40 × 15 × 5 mm^3) were installed under an argon atmosphere. The resulting mixture was stirred for 30 min without electricity and subsequently electrolyzed under a constant current of 9 mA at room temperature. After the reaction was complete as monitored by TLC after 12 h, the reaction mixture was transferred with ethyl acetate and quenched with a saturated aqueous solution of NH_4Cl (10 mL). The solid in the resulting mixture was dissolved with a minimal amount of water. The organic layer was separated, and the aqueous layer was extracted with ethyl acetate (20 mL × 3). The combined organic layer was washed sequentially with H_2O and brine, dried over anhydrous Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was purified by column chromatography to afford **3aa** (0.2272 g, 74%) (silica gel was pretreated with 5% (v/v) Et_3N in petroleum ether) [eluent: petroleum ether/ethyl acetate = 10/1 (1100 mL)] as a yellow oil.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.6c02017.

Detailed experimental procedures and characterization of new compounds (PDF)

Crystallographic information file for **3ka** (CIF)

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Notes

The authors declare no competing financial interest.

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